



## AI Playbook and Toolkit for Public Health Departments

# Human Factors, Workflow Integration, and Usability Testing

OPS 220

AI tools succeed or fail inside real public health workflows. Human factors and usability testing help agencies understand how staff actually use systems, where errors may occur, how cognitive load changes, and whether AI outputs support or distort decision-making. This module teaches learners to evaluate workflow fit, interface design, alert burden, documentation burden, and human review practices before deployment.

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| Level             | applied/foundational              |
| Primary track     | Operations and Data Quality Track |
| Appears in tracks | operations-data-quality           |

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## Learning Objectives

- Explain the purpose of human factors, workflow integration, and usability testing in public health AI work.
- Identify the technical, operational, privacy, security, equity, and governance issues associated with the topic.
- Apply the topic to a real or realistic public health workflow, system, or implementation scenario.
- Recognize common failure modes that can affect safety, accuracy, trust, workflow fit, or sustainability.
- Identify practical guardrails that should be documented before implementation.
- Produce a workflow usability test plan that can support readiness assessment, governance review, procurement, implementation, or monitoring.

## Module Overview

AI tools succeed or fail inside real public health workflows. Human factors and usability testing help agencies understand how staff actually use systems, where errors may occur, how cognitive load changes, and whether AI outputs support or distort decision-making. This module teaches learners to evaluate workflow fit, interface design, alert burden, documentation burden, and human review practices before deployment.

## Why This Topic Matters for Public Health AI

Human Factors, Workflow Integration, and Usability Testing matters because AI systems are embedded in public health programs, data systems, and decision processes rather than operating in isolation. The topic affects whether AI outputs are accurate, explainable, safe, equitable, secure, and sustainable. Agencies that ignore this area may create tools that work in a demonstration but fail in actual practice.

The topic also affects accountability. Public health agencies must be able to explain how data are used, why a tool was approved, what evidence supports the workflow, who reviews outputs, and what happens when the system fails. These responsibilities become more important as AI tools move from experimentation into operational use.

## Core Concepts

Human factors is the discipline of designing systems around human abilities, limits, environments, and tasks. In AI-supported public health work, human factors help agencies ensure that AI outputs are understandable, actionable, and reviewed appropriately.

Usability testing is structured observation of how users interact with a tool or workflow. It can reveal confusion, workarounds, over-trust, alert fatigue, documentation burden, or steps that are inconsistent with actual practice. Usability testing is a safety activity, not simply a design preference.

Cognitive load refers to the mental effort required to complete a task. AI can reduce burden by summarizing information, but it can also increase burden if staff must verify unclear outputs, interpret unexplained scores, or resolve conflicting recommendations.

## Public Health Example

A disease investigator may receive an AI-generated case summary in the surveillance system. If the summary hides uncertainty or makes unsupported claims look official, the investigator may over-trust it. If the summary is too long, poorly organized, or disconnected from source fields, it may not reduce burden. Usability testing would observe investigators using the summary and identify changes needed before launch.

## Technical Deep Dive

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Workflow integration begins with current-state mapping. Agencies should document how work happens today, including informal workarounds, duplicate entry, handoffs, delays, and decision points. AI should be introduced only after the team understands which step is being improved and how the change affects users.

Future-state design should define how staff will see, review, correct, accept, reject, or escalate AI outputs. The user interface should make source information available, show uncertainty when appropriate, and avoid presenting AI outputs as final decisions. Good design supports human judgment rather than bypassing it.

Usability testing should include realistic users, realistic tasks, and realistic data. Testing should evaluate not only whether users can complete a task, but also whether they understand the AI output, recognize limitations, know when to escalate, and can document their final decision.

## **Risks, Failure Modes, and Guardrails**

A common failure mode is treating the topic as a technical detail rather than a public health control. When staff do not connect technical decisions to authority, workflow, equity, privacy, security, and accountability, the agency may adopt a tool that appears useful but cannot be sustained or defended.

Another risk is relying on vendor claims without agency-defined evidence. Vendors may provide demonstrations, dashboards, or summary metrics that do not match the agency workflow or data environment. A guardrail is to require documentation, test results, acceptance criteria, and agency-controlled review.

A third risk is failing to plan for change. Public health data, policies, systems, and emergencies change. Any technical approach must include monitoring, review, versioning, escalation, and retirement pathways so the agency can respond when assumptions no longer hold.

## **Application to AI-Supported Workflows**

Generative AI, predictive analytics, RAG, automation, and agentic workflows all depend on the controls described in this module. The controls should be scaled to the risk of the use case. A tool that drafts internal training text may require lighter review than a tool that updates case records, prioritizes outreach, or supports public communication during an emergency.

The module should be applied during readiness assessment, procurement, implementation planning, pilot testing, and post-deployment monitoring. Learners should document how the topic affects data, systems, users, safeguards, and decision-making for the selected workflow.

## **Reflection Questions**

Where does this topic appear in one current or proposed AI workflow in your agency?

What decisions, data sources, users, or partners would be affected if this area is poorly designed?

What evidence would governance reviewers need before approving the workflow?

Which safeguards should be required before pilot testing?

Who owns ongoing monitoring and review?

## **Practical Exercise**

Create a workflow usability test plan for one real or realistic public health AI workflow. The artifact should describe the use case, affected users, data sources, system dependencies, risks, guardrails, review points, and unresolved questions.

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The exercise should produce a work product that could be used in a governance meeting, procurement review, implementation planning session, pilot evaluation, or monitoring review. Learners should document assumptions and identify which staff or partners need to be consulted.

## Expected Artifact or Evidence

The expected artifact is a completed workflow usability test plan. It should be specific to a public health workflow and should include technical dependencies, governance questions, human review expectations, monitoring indicators, and unresolved risks.

## Expected Assignment

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## Knowledge Check

- 1. What is the best reason to address human factors, workflow integration, and usability testing before deployment?
- 2. Which practice best supports responsible implementation?
- 3. Which statement best describes a common failure mode?
- 4. What should be included in the practical artifact for this module?
- 5. When should the issue be escalated for governance or leadership review?

## References and Resources

- NIST AI Risk Management Framework: <https://www.nist.gov/itl/ai-risk-management-framework>
- AHRQ human factors and usability resources
- NIST Generative AI Profile: <https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence>
- CDC Public Health Data Strategy and Data Modernization resources: <https://www.cdc.gov/public-health-data-strategy/php/about/index.html>
- OWASP Top 10 for LLM Applications, when security or AI integrations are relevant
- Agency privacy, cybersecurity, data governance, procurement, and records management policies